CompSci 469/469G
Intro to Computer Security
Cybersecurity Foundations

Administrivia

- Setup handouts available on canvas
- Must get the Textbook
- Class activity Next Thursday [Sept 11]
- My office hours:
  - Tuesday, Thursday 2.30-3.30pm (EMS 817)

Why Security of a System Matters?





Gmail Security Warning are False

CIA (Confidentiality, Integrity, Availability)

- **Confidentiality**: Protecting data from unauthorized access.
- **Integrity**: Ensuring information isn't tampered with.
- **Availability**: Making sure systems are accessible when needed.



Confidentiality

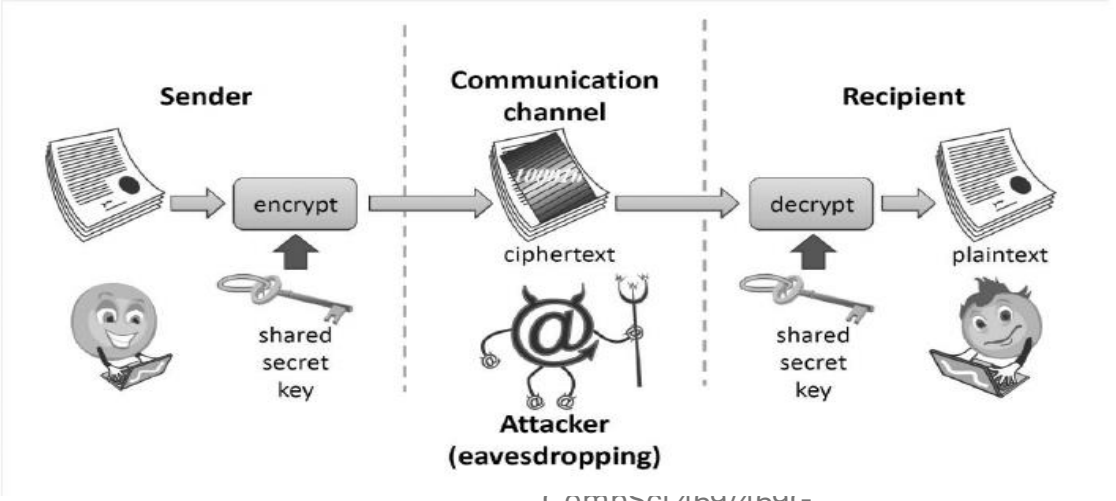
- Confidentiality Prevents unauthorized disclosure of information

e.g., only authorized payroll employees should have access to the employee payroll database.



Tools for Confidentiality

- **Encryption** is the process of converting information into an unreadable form using a secret value – an encryption key.
- The transformed information can only be restored with a decryption key, in some cases, both keys are same.



Tools for Confidentiality

- **OpenSSL -** A widely used library and tool for encryption
- Supports:
  - Symmetric & public/private key encryption
  - Digital signatures & certificates
  - Hash functions (e.g., SHA-256)
  - Random number generation



Tools for Confidentiality

- **Access Control -** *rules and policies that limit access to confidential information to right users*
- In Linux:
  - /etc/passwd file - readable by everyone
  - /etc/shadow (stores encrypted passwords) - restricted only to the rot user.

```
-rw-r--r--.  1 root root     2640 Aug 10 18:43 passwd
-rw-r--r--.  1 root root     2640 Aug 10 18:43 passwd-
----------.  1 root root     1393 Aug 10 18:59 shadow
----------.  1 root root     1179 Aug 10 18:43 shadow-
```

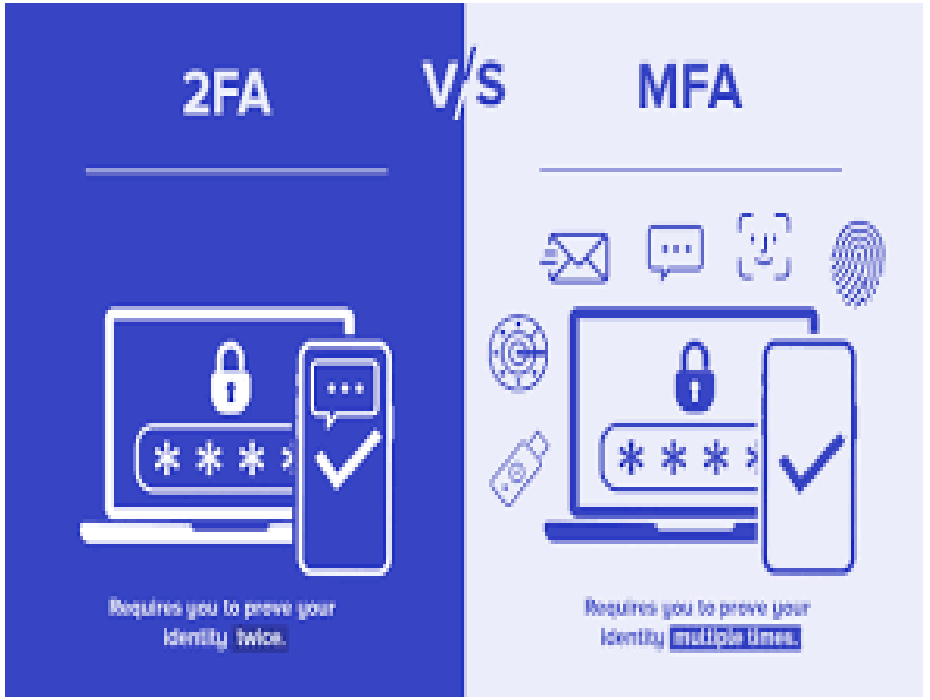
Tools for Confidentiality

- **Authentication** is the process of verifying a user's identity confirming that someone really is who they claim to be.
- Common Authentication Factors:
  - Something you know
  - Something you have
  - Something you are

Example?

Tools for Confidentiality

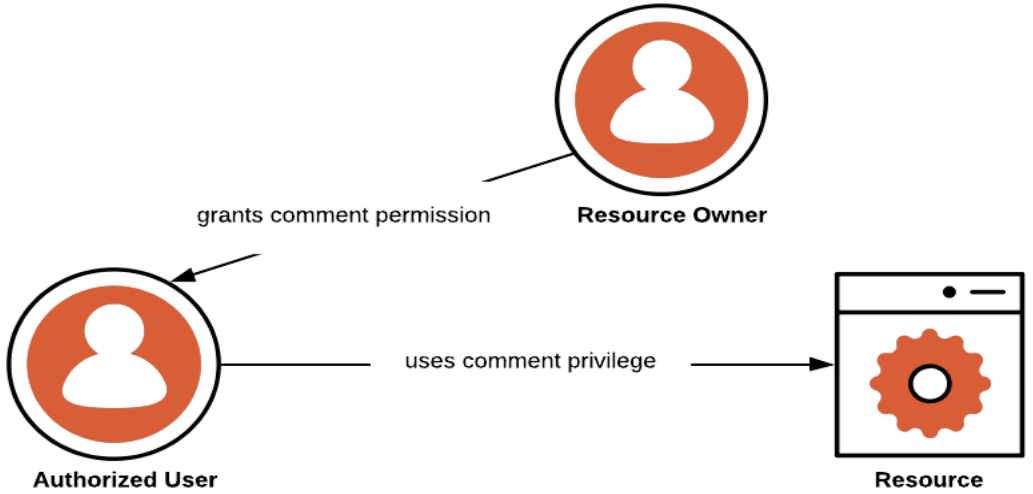
- **Authentication** is the process of verifying a user's identity confirming that someone really is who they claim to be.
- Common Authentication Factors:
  - Something you know
    e.g. Password, or security question
  - Something you have
    e.g. smart card, or phone app
  - Something you are
    e.g. fingerprint, face, voice, palm



Tools for Confidentiality

- **Authorization** is the process of deciding what actions or resources a user is allowed to access after they have been authenticated.

  e.g. student can view their grades but cannot change them.



Tools for Confidentiality

- **Physical Security** – barriers that protect systems from physical access

  e.g. Server rooms locked and monitored with cameras

Integrity

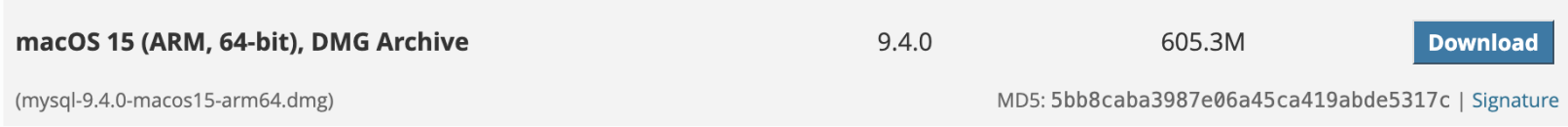
- Integrity means that information has not been changed in an unauthorized way.
- When Integrity Can Be Violated
  - Hackers modify configuration files to hide their tracks.
  - Someone changes system logs to cover evidence.
  - Human error: accidentally overwriting or deleting files.

- Tools:
  - Backups: the periodic archiving of data.
  - Checksums (hash functions): Create a unique number for a file; if it changes, the file was altered.
  - Digital signatures: method ensures that the information originated from the signer and was not altered

Integrity

- Hash function example:

On the site https://dev.mysql.com/downloads/mysql



- before installing compare the MD5 checksum of your downloaded file with the one provided by the website. If they match, the file is intact and has not been corrupted or tampered with.

Availability

- Availability - Information must be accessible and usable on time by authorized people.
- Denial of Service (DoS) attack → overloads a system so it becomes unavailable.
- Tools:
- Physical Protections → backup power (UPS), fire suppression, secure server rooms.
- Computational Redundancy → backup servers, mirrored storage.



Next Class

Chapter 1 – Linux Security Basics

TODO: Learn these commands from man page

- `pwd`
- `ls`
- `cd`
- `echo`
- `cat`
- `cp`